

Design of a Hamiltonian-Based Vision Kernel (HVK)

using a Hybrid Architecture and Tensor Network Framework

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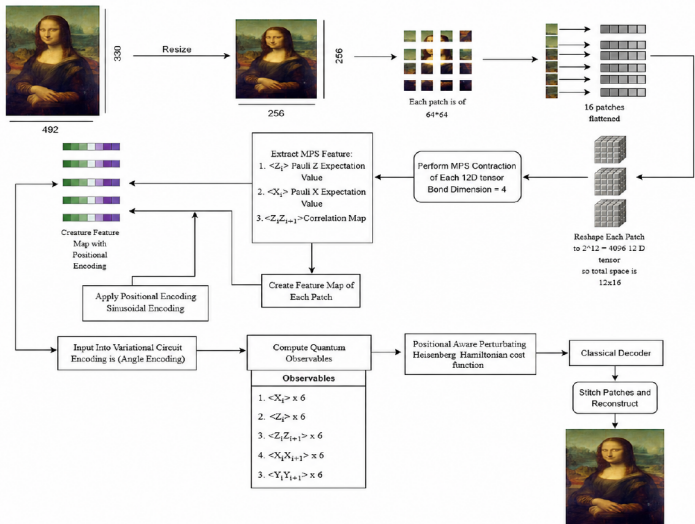
Overview

1. Motivation & Idea
2. Pipeline & Image Preparation
3. Results: Reconstruction Demos
4. Sanity Checks & Diagnostics
5. Quantitative Results
6. Does the Quantum Part Actually Help?
7. Conclusion

Motivation & Idea

- Quantum models for high-definition images: too many qubits, too much circuit depth.
- High-dimensional data needs compression *without* losing correlation and fidelity.
- **Our approach:** patch decomposition $\rightarrow$ MPS compression $\rightarrow$ quantum feature extraction $\rightarrow$ classical decoder.
- A learnable Heisenberg energy term adds a physics-inspired inductive bias.

Model Pipeline



Positional Encoding

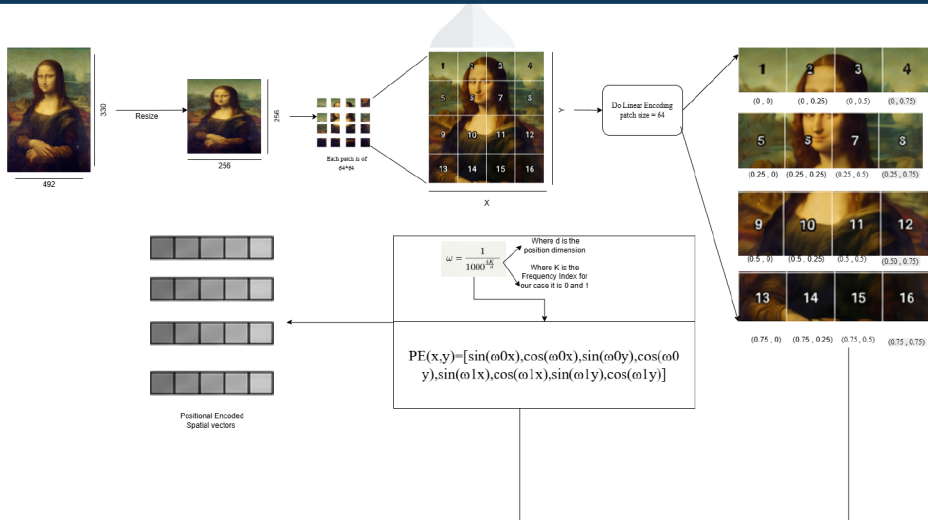


Image Reconstructed Without the Hamiltonian Kernel

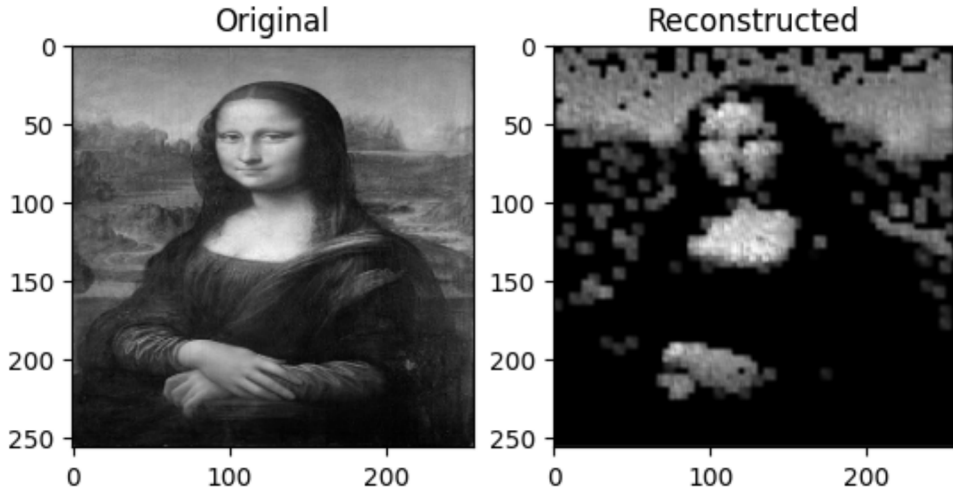


Image Reconstructed using Only positional field Encoding

Quantum Field Reconstruction



Image Created with Hamiltonian Vision Kernel



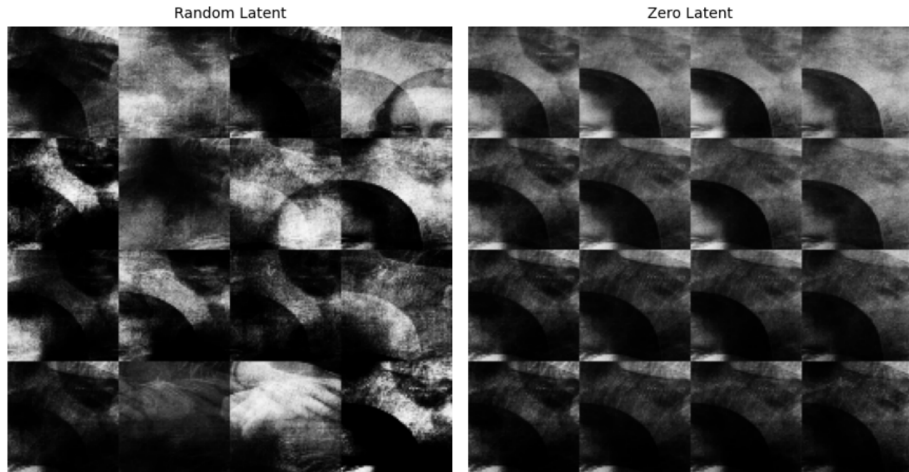
Original



Quantum Recon



random latent and zero latent to check if decoder is cheating!!

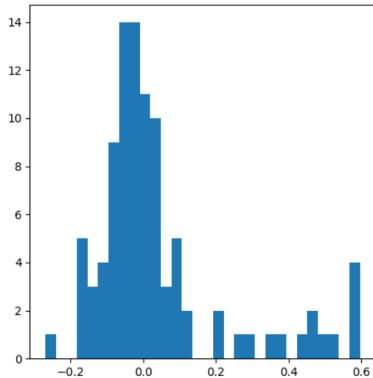


As zero latent and random latent is bogus this proves that decoder is highly latent dependent

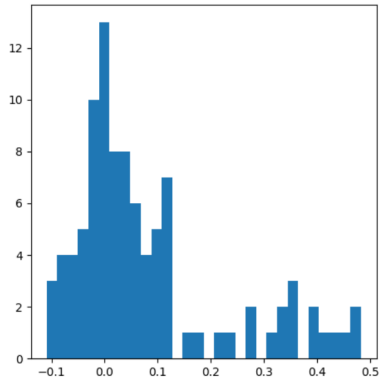
Observable Distribution



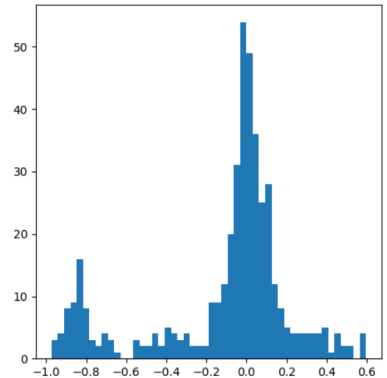
Pauli-Z Distribution



Pauli-X Distribution

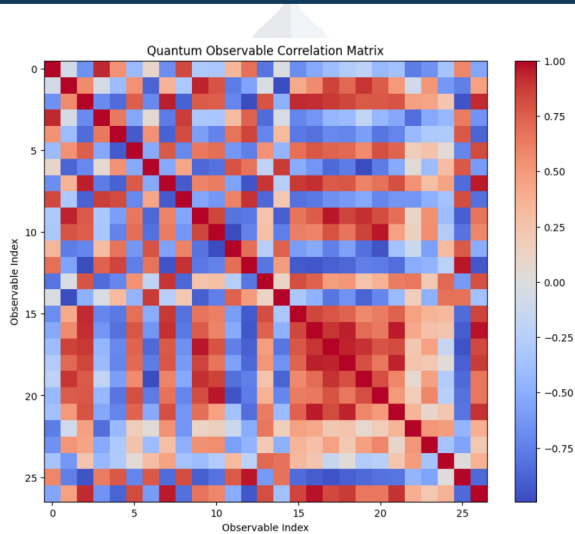


All Observable Distribution

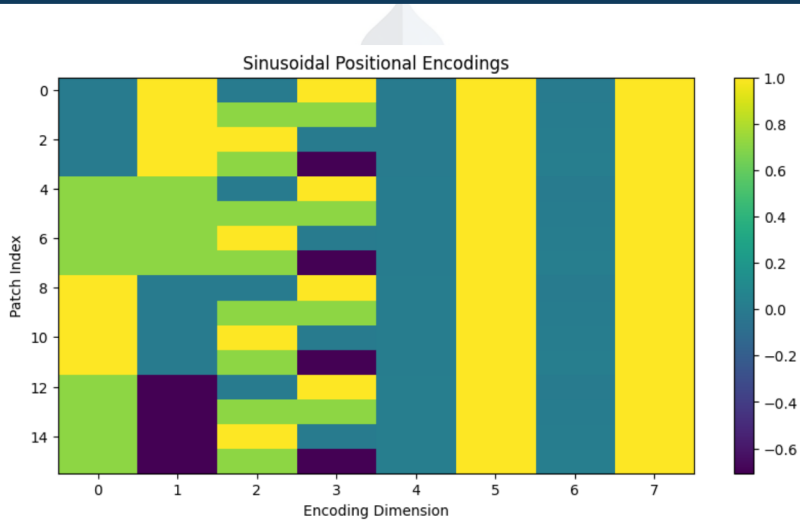


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Correlation Matrix



Positional Encoding Heat Map



Quantitative Results (Monalisa Demo)

Metric	Value
MSE	1.953763×10^{-3}
MAE	3.166284×10^{-2}
PSNR	27.09128 dB
Total Parameters	1.090691×10^6
Model Parameters	387
Decoder Parameters	1.090304×10^6
Observable Mean	-1.053798×10^{-1}
Observable Std	3.428020×10^{-1}
Mean Entropy	9.944191×10^{-3}
Max Entropy	1.364326×10^{-1}
Mean Energy	-2.525678
Std Energy	1.780927×10^{-1}

Does the Quantum Part Actually Help?

- The demo above shows HVK *reconstructs* images well.
- It does not show *which component* is responsible.
- So we ran a resource-matched, freeze-based, multi-seed ablation study.

This is what we found.

What We Found: The Classical Decoder Does the Work

Variant (Monalisa, 5 seeds)	PSNR (dB)	Verdict
Freeze classical, train quantum	10.93 ± 0.00	Collapses
Shared VQC baseline	32.97 ± 0.22	Reference
No entanglement / No MPS	≈ 33.0	Not significant
Classical tanh replacement	33.50 ± 0.00	Exceeds baseline

Held-out CIFAR-10: raw-linear (18.80 ± 1.42 dB) beats trained HVK2D (18.12 ± 1.54 dB), $p = 9.5 \times 10^{-6}$.

- We also caught and withdrew a leakage bug in an early diagnostic (a too-good-to-be-true $R^2 = 1.0$).
- Two narrow positive results survive: entanglement is necessary on a synthetic pair-correlation task, and a D_4 -pooled variant is provably symmetric.

Conclusion

- HVK reconstructs images well and its representation is physically interpretable.
- But on held-out real images, the classical decoder — not the quantum circuit — is load-bearing.
- We report this honestly, because the evaluation protocol is the real contribution.

THANK YOU!

